

BUSINESS ANALYSIS PROJECT

E-Commerce Order Management System

Karoo Market

Industry: E-commerce

Role: Business Analyst

Project Type: Digital Process & System Improvement

Portfolio Case Study

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1. Business Case

1.1 Project Overview

Karoo Market is an e-commerce business experiencing operational issues across its order lifecycle. Orders are currently processed through disconnected systems, creating delays, inaccurate inventory information, payment-related failures, limited order visibility, and avoidable customer complaints.

1.2 Business Problem

The current order-management process does not provide a single, reliable view of orders, inventory, payments, and fulfilment status. Manual processing and system fragmentation increase the risk of incorrect stock information, duplicate orders, delayed fulfilment, and poor customer communication.

1.3 Business Objectives

- Centralise order information and processing.
- Improve inventory accuracy and availability visibility.
- Integrate payment processing into the order lifecycle.
- Provide customers with reliable order tracking.
- Reduce manual order-processing activities.
- Reduce order-related complaints and processing delays.

1.4 Current Performance Baseline

Metric	Current	Target
Order processing time	14 minutes	< 5 minutes
Inventory accuracy	82%	98%
Payment failure rate	8%	< 3%
Order-related complaints	16%	< 7%
Manual processing	65%	< 20%

2. Stakeholder Analysis

Stakeholders are individuals or groups affected by the order-management process or involved in implementing the solution.

Stakeholder	Role / Interest	Influence	Key Need
Customers	Purchase products and track orders	High	Accurate orders and visibility
Sales Team	Supports sales and customer orders	High	Reliable order status
Warehouse	Picks, packs and dispatches orders	High	Accurate inventory
Finance	Handles payments, refunds and reconciliation	High	Reliable payment data
Customer Service	Handles order-related queries	High	Real-time order information
IT Team	Implements and supports	High	Clear technical

	solution		requirements
Management	Owens business outcomes	High	Efficiency and revenue improvement
Delivery Partners	Fulfil customer deliveries	Medium	Accurate shipment details

3. Problem Statement & Scope

3.1 Problem Statement

Karoo Market's fragmented order-management process creates inconsistent information between sales, inventory, payment, warehouse and customer-service activities. This leads to slow processing, incorrect stock visibility, payment failures, duplicate orders and poor customer experience.

3.2 In Scope

- Customer order creation and checkout.
- Inventory validation and update.
- Payment processing integration.
- Order confirmation and notifications.
- Order-status tracking.
- Order cancellation and refund initiation.
- Warehouse order visibility.
- Operational reporting.

3.3 Out of Scope

- Product manufacturing.
- Supplier procurement systems.
- Payroll and HR systems.
- Redesign of the company's entire accounting platform.

4. Requirements

4.1 Functional Requirements

ID	Requirement	Priority
FR-001	The system must allow customers to add products to a shopping cart.	High
FR-002	The system must validate product availability before order confirmation.	Critical
FR-003	The system must calculate the order total, including applicable delivery charges.	High
FR-004	The system must process payments through an approved payment	Critical

	provider.	
FR-005	The system must create an order after successful payment.	Critical
FR-006	The system must update inventory after a successful order.	Critical
FR-007	The system must generate order confirmation.	High
FR-008	Customers must be able to view order status.	High
FR-009	Customers must be able to request order cancellation where business rules permit.	Medium
FR-010	The system must support refund initiation for eligible cancelled orders.	High
FR-011	Warehouse staff must be able to view orders requiring fulfilment.	High
FR-012	Management must be able to access order and operational performance reports.	Medium

4.2 Non-Functional Requirements

ID	Requirement	Priority
NFR-001	Checkout pages should respond within 3 seconds under normal operating conditions.	High
NFR-002	Customer payment information must be protected using appropriate security controls.	Critical
NFR-003	The solution should be available 99.5% of the time.	High
NFR-004	The system must maintain an audit trail for key order-status changes.	High
NFR-005	The solution must support desktop and mobile users.	High
NFR-006	The system should support concurrent users during peak sales periods.	High

5. User Stories & Acceptance Criteria

US-001 — Place an Order

As a customer, I want to place an order online so that I can purchase products without contacting customer service.

Acceptance Criteria:

- Customer can add an available product to the cart.
- System validates stock before order confirmation.
- Customer can select a valid payment method.
- Successful payment creates an order.
- Customer receives an order confirmation.

US-002 — Track an Order

As a customer, I want to track my order so that I know its status and expected delivery.

Acceptance Criteria:

- Customers can view the order from their account.
- Current order status is displayed.
- Estimated delivery date is displayed where available.
- Tracking information is displayed when provided by the delivery partner.

US-003 — Update Inventory

As a warehouse employee, I want inventory to update automatically when an order is completed so that stock levels remain accurate.

Acceptance Criteria:

- Inventory decreases after successful order creation.
- Unavailable products cannot be oversold.
- Updated stock is visible to authorised users.

US-004 — Process Payment

As a customer, I want my payment to be processed securely so that my order can be confirmed.

Acceptance Criteria:

- Payment request is sent to the approved payment provider.
- Successful payment returns a success response.
- Failed payment does not create a confirmed order.
- Customer receives appropriate payment feedback.

US-005 — Manage Fulfilment

As a warehouse employee, I want to see paid orders requiring fulfilment so that I can pick, pack and dispatch them.

Acceptance Criteria:

- Paid orders appear in the fulfilment queue.
- Order details and quantities are visible.
- Order status can be updated as fulfilment progresses.

6. Business Rules

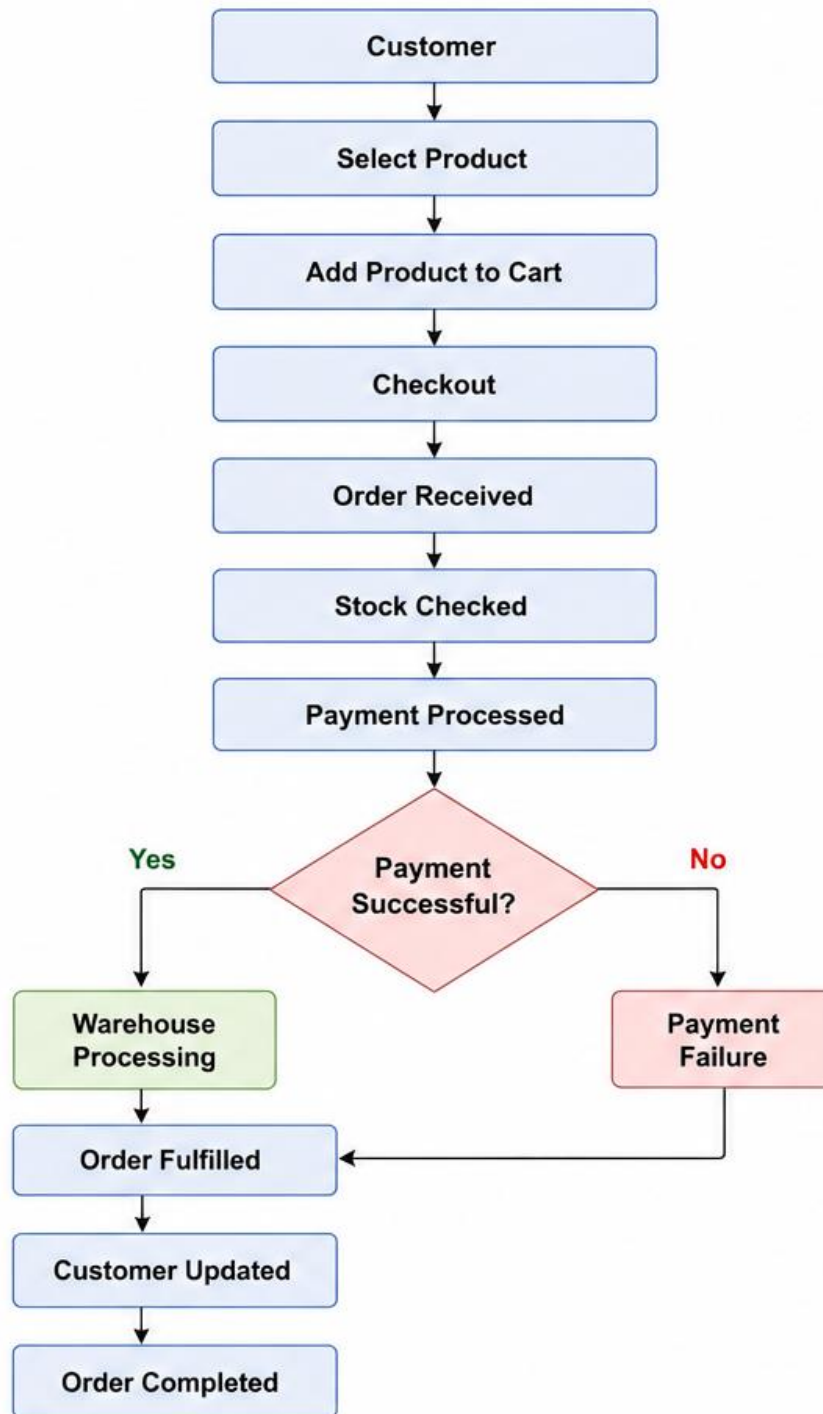
ID	Business Rule
BR-001	An order cannot be confirmed unless required customer and delivery information is provided.
BR-002	A product must have sufficient available stock before it can be ordered.
BR-003	A confirmed order must have a successful payment response or an approved alternative payment status.
BR-004	Inventory must be updated when an order is successfully confirmed.
BR-005	A cancelled order may only be refunded according to the applicable refund policy.
BR-006	Orders already dispatched cannot be cancelled through the standard customer workflow.
BR-007	Only authorised staff may change order status manually.
BR-008	Every material order-status change must be recorded in the audit trail.

7. Order Management Process

7.1 As-Is Process

Customer places an order → Sales/order system receives order → Stock is checked manually or through a separate system → Payment is processed → Warehouse receives order → Order is fulfilled → Customer is updated.

Figure 1: As-Is Order Management Process

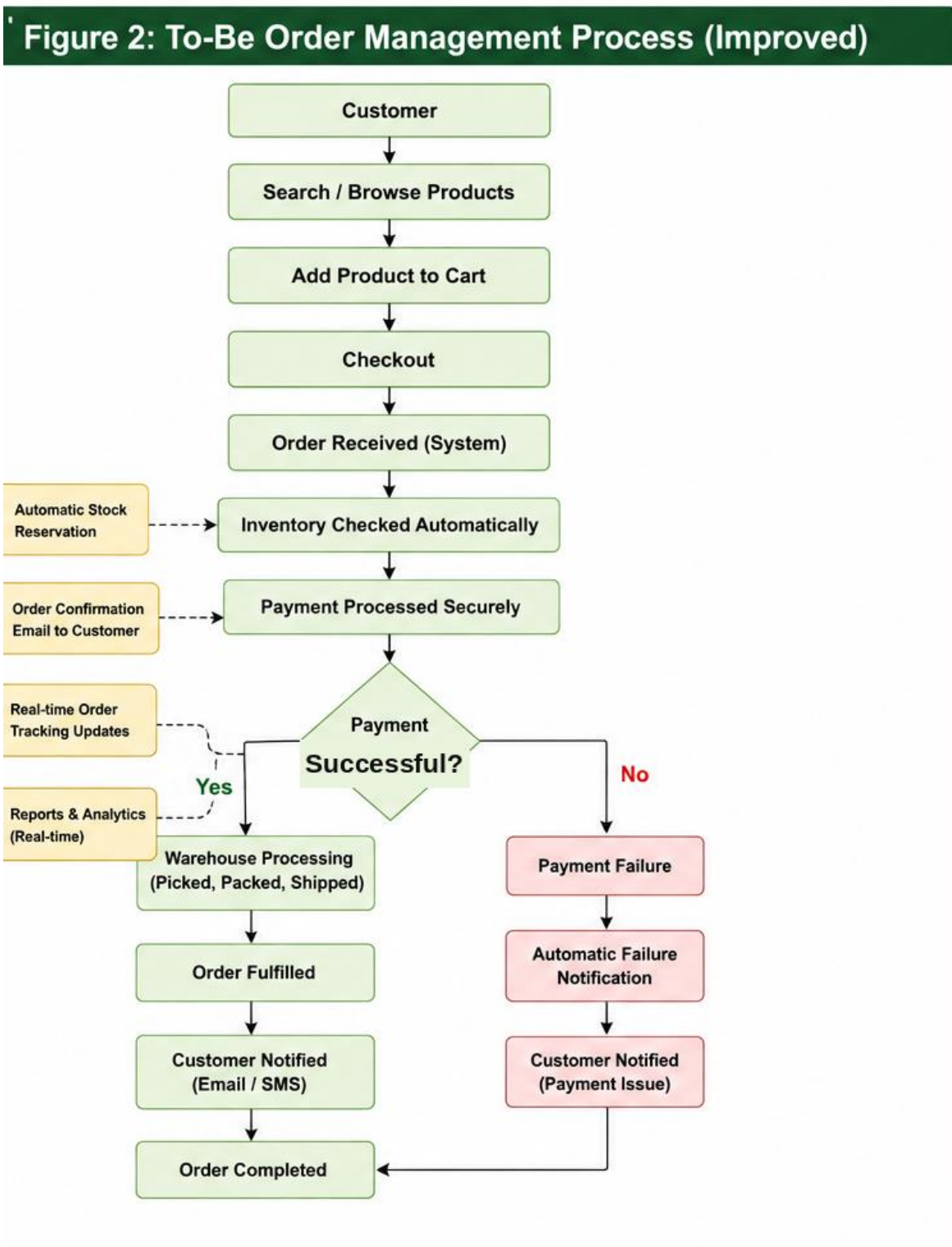


7.2 Key As-Is Problems

- Disconnected systems create inconsistent order information.
- Inventory information may not be updated immediately.
- Payment and order confirmation are not fully integrated.
- Customers have limited real-time order visibility.
- Manual intervention increases processing time and errors.

7.3 To-Be Process

Customer selects product → System validates stock → Customer checks out → Payment is processed → Order is created automatically → Inventory is updated → Confirmation is sent → Warehouse receives fulfilment request → Order is dispatched → Tracking is updated → Customer receives status notifications.



7.4 Process Improvement Summary

Area	Current Process	Proposed Process
Order creation	Multiple/manual steps	Centralised
Stock validation	Disconnected/manual	Real-time validation
Payment	Separate process	Integrated
Inventory update	Delayed risk	Automatic
Customer tracking	Limited	Self-service tracking
Notifications	Manual/inconsistent	Automated
Reporting	Fragmented	Centralised reporting

8. Use Case — Place an Order

Field	Description
Use Case ID	UC-001
Use Case	Place an Order
Primary Actor	Customer
Supporting Actors	Payment Provider, Inventory Service, Warehouse
Preconditions	Products are available and customers have provided required checkout information.
Trigger	Customer selects 'Place Order'.
Postcondition	Confirmed order exists and inventory is updated after successful payment.

8.1 Main Flow

- Customer reviews cart.
- System validates stock.
- System calculates order total.
- Customer submits payment.
- Payment provider returns successful payment response.
- System creates confirmed order.
- System updates inventory.
- System sends order confirmation.
- Warehouse receives fulfilment requests.

8.2 Alternative Flows

- If stock is unavailable, the customer is informed and the order cannot be confirmed for that item.
- If payment fails, the order remains unconfirmed and the customer is prompted to retry or select another payment method.
- If a technical failure occurs after payment, the transaction must be reconciled before the order is confirmed or refunded.

9. Data Analysis & KPIs

The BA should use operational data to validate the problem, establish a baseline and monitor the solution after implementation.

KPI	Current	Target	Business Purpose
Order processing time	14 min	< 5 min	Measure operational efficiency
Inventory accuracy	82%	98%	Reduce stock-related issues
Payment failure rate	8%	< 3%	Improve successful checkout
Order-related complaints	16%	< 7%	Improve customer experience
Manual processing	65%	< 20%	Reduce operational workload
Order fulfilment SLA	78%	95%+	Improve delivery readiness
Order cancellation rate	9%	< 6%	Identify avoidable order issues

10. Test Cases

Test ID	Requirement	Test Scenario	Expected Result
TC-001	FR-001	Add an available product to cart	Products appear in carts with correct quantity.
TC-002	FR-002	Attempt to purchase an out-of-stock product	System prevents confirmation and displays availability message.
TC-003	FR-003	Complete checkout with delivery charges	Correct total is calculated.
TC-004	FR-004	Submit valid payment	Payment succeeds and response is recorded.
TC-005	FR-005	Complete successful payment	Confirmed order is created with unique order number.
TC-006	FR-006	Complete successful order	Inventory quantity is reduced correctly.
TC-007	FR-007	Complete successful order	Customer receives confirmation.
TC-008	FR-008	Open an existing order	Status and delivery information are displayed.
TC-009	FR-009	Cancel an eligible order	Cancellation requests are accepted according to policy.
TC-010	FR-010	Process eligible refund	Refund requests are initiated and recorded.
TC-011	FR-011	Warehouse opens fulfilment queue	Paid orders requiring fulfilment are displayed.
TC-012	FR-012	Manager requests performance report	Relevant order KPIs are generated.

11. Requirements Traceability Matrix

Requirement ID	Requirement	User Story	Test Case	Priority
FR-001	Shopping cart	US-001	TC-001	High
FR-002	Stock validation	US-001	TC-002	Critical
FR-003	Order total calculation	US-001	TC-003	High
FR-004	Payment processing	US-004	TC-004	Critical
FR-005	Order creation	US-001	TC-005	Critical
FR-006	Inventory update	US-003	TC-006	Critical
FR-007	Order confirmation	US-001	TC-007	High
FR-008	Order tracking	US-002	TC-008	High
FR-009	Order cancellation	US-001	TC-009	Medium
FR-010	Refund initiation	US-001	TC-010	High
FR-011	Fulfilment queue	US-005	TC-011	High
FR-012	Operational reporting	—	TC-012	Medium

12. Risks & Mitigations

Risk	Severity	Potential Impact	Mitigation
Incorrect inventory data	High	Overselling and customer complaints	Real-time stock validation and reconciliation
Payment integration failure	High	Orders cannot be confirmed	Error handling, retry logic and payment reconciliation
Poor user adoption	Medium	Staff continue manual workarounds	Training, stakeholder involvement and monitoring
System downtime	High	Lost sales and customer disruption	Availability monitoring and recovery procedures
Data security breach	Critical	Customer and financial impact	Access controls, security testing and audit logging

13. Final Recommendation

13.1 Recommended Solution

Karoo Market should implement a centralised Order Management System that integrates order creation, inventory validation, payment processing, fulfilment, customer notifications and order tracking.

13.2 Implementation Priorities

- Phase 1: Centralise order creation, stock validation and payment integration.
- Phase 2: Integrate warehouse fulfilment and customer order tracking.

- Phase 3: Implement notifications, dashboards and continuous KPI monitoring.

13.3 Expected Business Benefits

- Faster order processing.
- Higher inventory accuracy.
- Fewer payment-related order failures.
- Reduced manual processing.
- Improved customer visibility and satisfaction.
- Better operational reporting and decision-making.

13.4 Conclusion

The proposed Order Management System addresses the main process and information gaps identified in Karoo Market's current environment. By centralising order data and automating key activities, the business can improve operational efficiency, reduce avoidable errors and provide customers with more reliable purchasing experience.

14. Portfolio Summary

This case study demonstrates practical Business Analyst capabilities including business problem analysis, stakeholder analysis, requirements engineering, user stories, acceptance criteria, business rules, process analysis, use cases, KPI definition, test-case design, requirements traceability, risk analysis and solution recommendation.

Suggested portfolio title: Karoo Market — E-Commerce Order Management System Business Analysis Case Study.